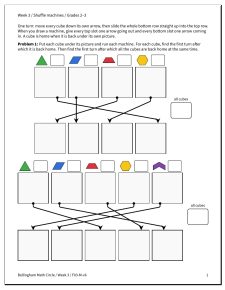
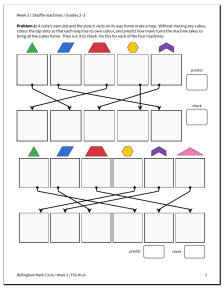
Week 3 / Shuffle machines / Problem menu

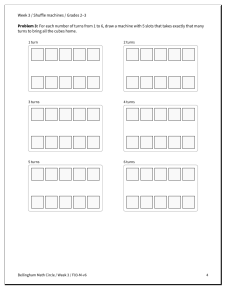
for adults: pick, then print



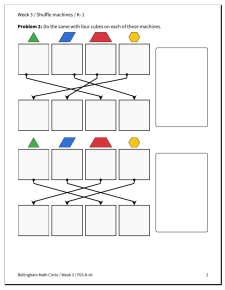
Run each machine until every cube is home
K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1
first route
wording differs by band



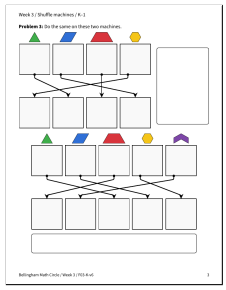
Predict the turns without moving cubes
Grades 2–3 P2 · Grades 4–5 P2
first route
wording differs by band



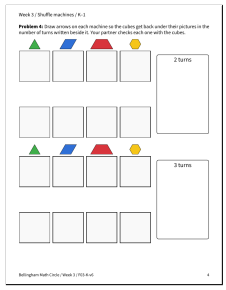
Turn counts with 5 slots
Grades 2–3 P3 · Grades 4–5 P3
first route
wording differs by band



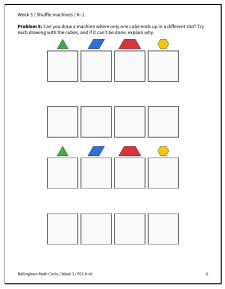
Four cubes on two machines
K–1 P2
first route



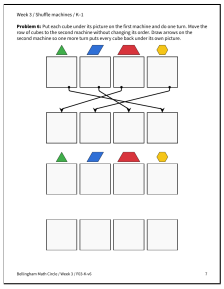
A 4-slot and a 5-slot machine
K–1 P3
first route



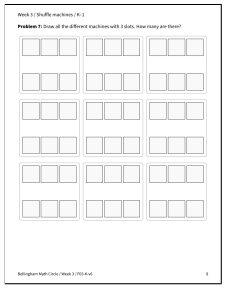
Arrows for 2, 3 and 4 turns
K–1 P4
first route



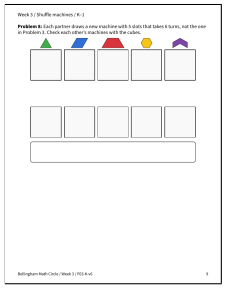
Only one cube out of place
K–1 P5
return route



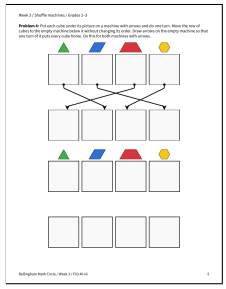
Draw the machine that sends every cube home
K–1 P6
return route



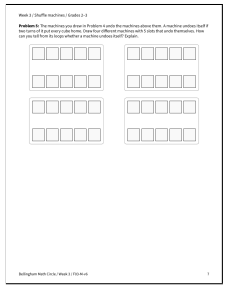
All the machines with 3 slots
K–1 P7
return route



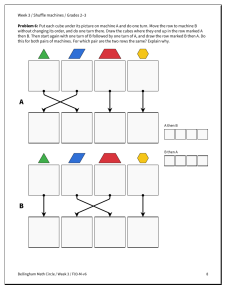
A 5-slot machine that takes 6 turns
K–1 P8
return route



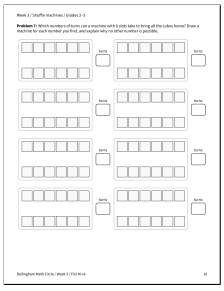
Draw the machine that undoes one turn
Grades 2–3 P4
first route



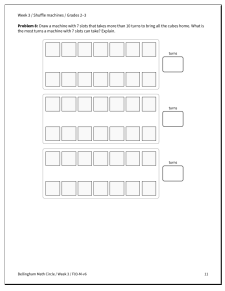
Machines that undo themselves
Grades 2–3 P5
return route



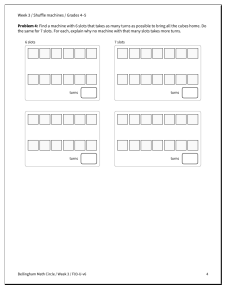
A then B and B then A
Grades 2–3 P6
return route



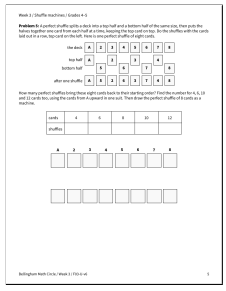
Turn counts with 6 slots
Grades 2–3 P7
return route



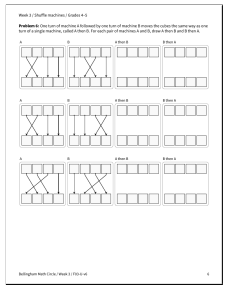
More than 10 turns with 7 slots
Grades 2–3 P8
return route



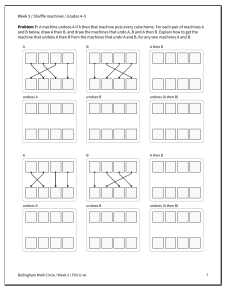
Slowest machines with 6 and 7 slots
Grades 4–5 P4
first route



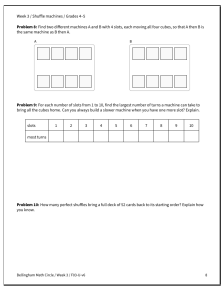
Perfect shuffles of 4 to 12 cards
Grades 4–5 P5
first route



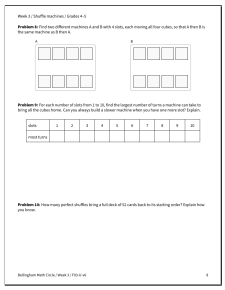
A then B and B then A for three pairs
Grades 4–5 P6
return route



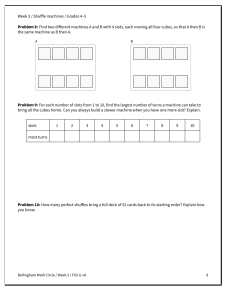
Undoing A then B
Grades 4–5 P7
return route



Two machines with A then B equal to B then A
Grades 4–5 P8
return route



Slowest machines for 1 to 10 slots
Grades 4–5 P9
return route



Perfect shuffles of a full deck
Grades 4–5 P10
return route